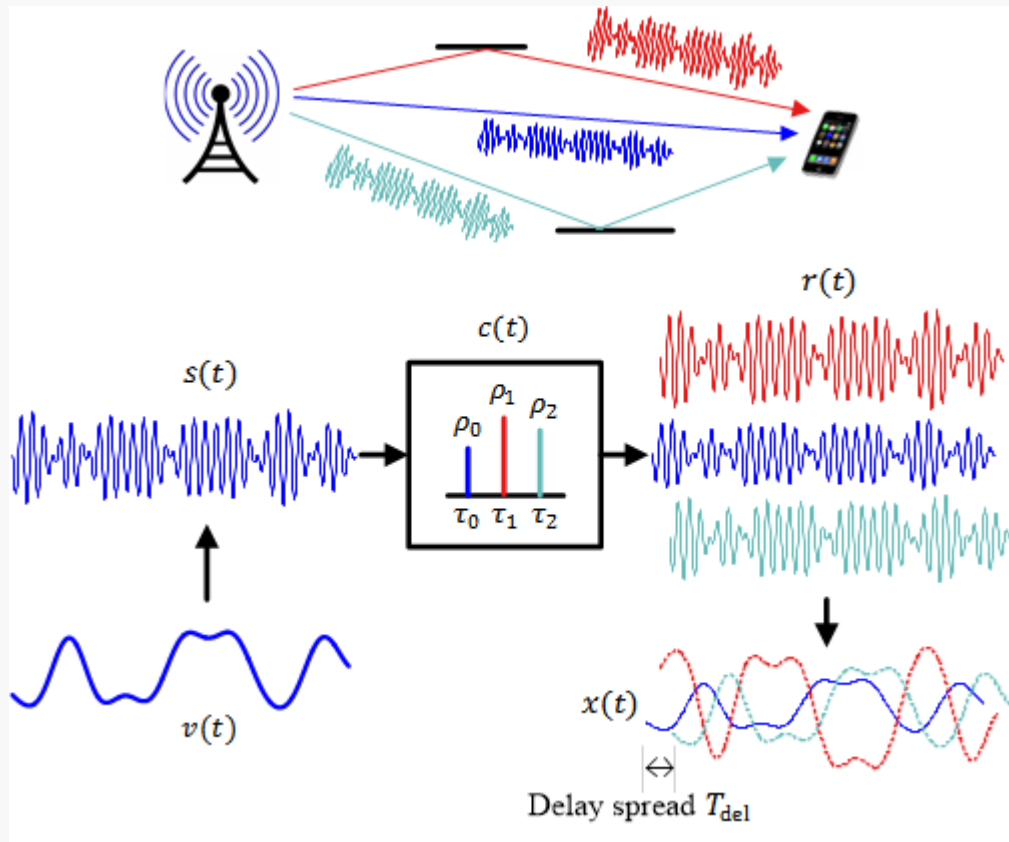


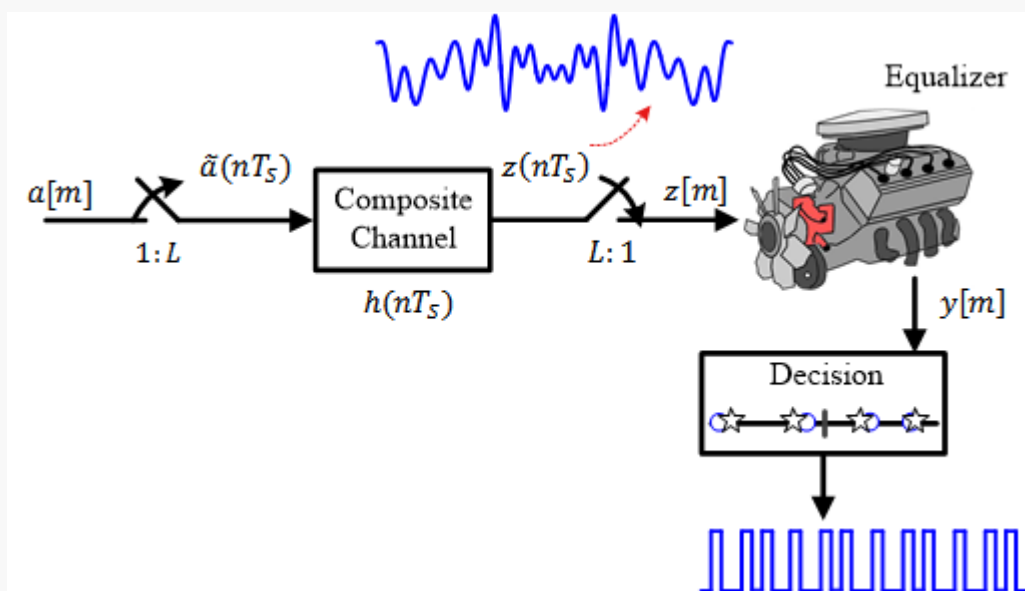
Equalization

Qasim Chaudhari



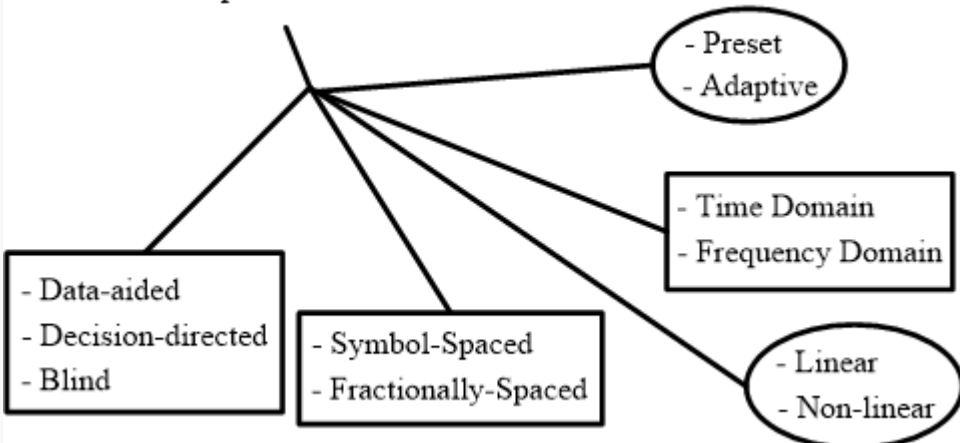
Equalization

- Any signal processing technique that is designed to eliminate or reduce the ISI before symbol detection

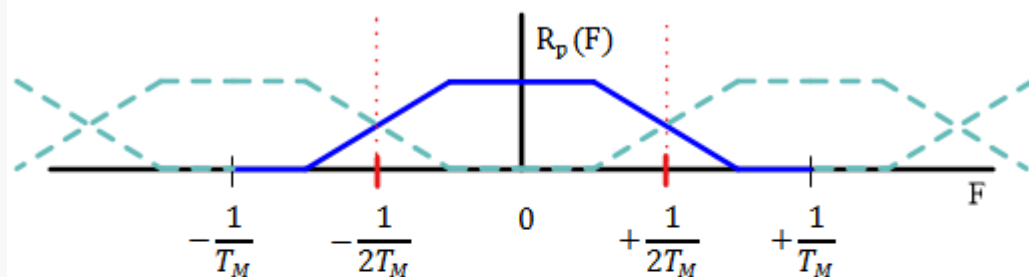


The Big Picture

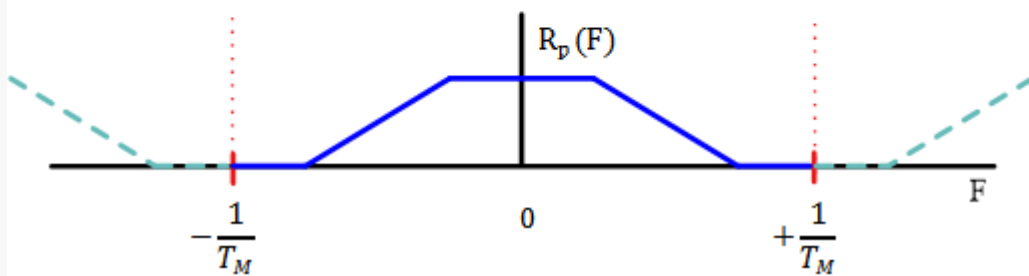
Equalizers



Input to a T_M -Spaced Equalizer



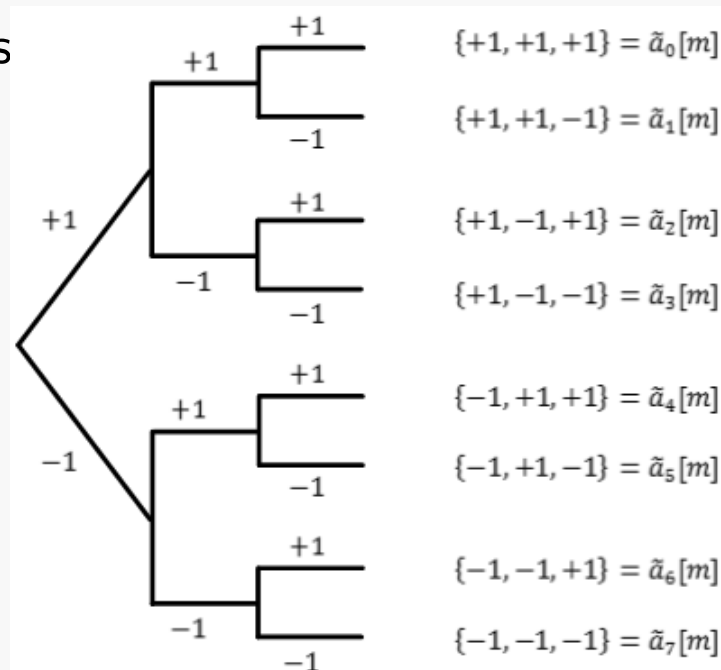
Input to a $T_M/2$ -Spaced Equalizer



Brute Force Equalization

Known as Maximum Likelihood Sequence Estimation

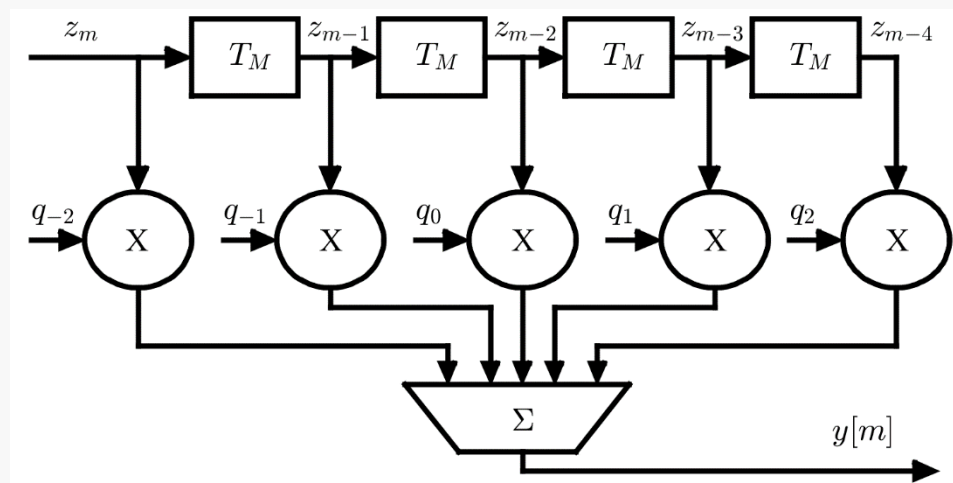
- Modulation is finite set of size M (BPSK: $M = 2$, QPSK: $M = 4$)
- Length 3 BPSK sequence $\rightarrow 2^3 = 8$ possible output sequences
- Convolve the channel with all possible options
- Choose the one with minimum error



Linear Equalizers

Employ an FIR digital filter $q[m]$ as an equalizer at the Rx with $2L_q + 1$ taps

- Extra 1 tap refers to the main tap
- Input is the matched filter output $z[m]$
- Output is the best (in terms of a chosen criterion) estimate of the Tx symbol $y[m]$ (sliced to $\hat{a}[m]$)



Zero Forcing Equalizer

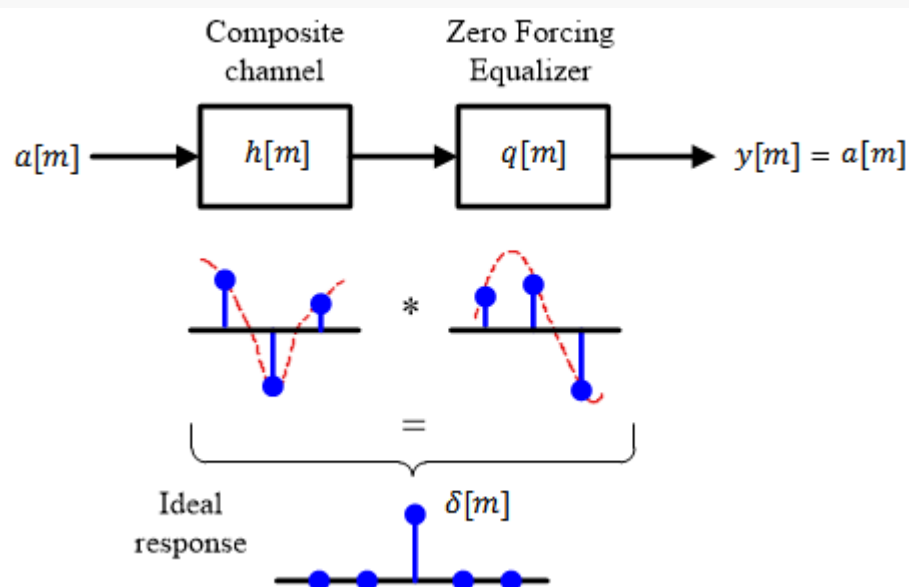
Main idea: Composite channel does not satisfy the Nyquist no-ISI criterion

$$h[m] \neq \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases}$$

- Expand the equalizer output $y[m] = z[m] * q[m] = a[m] * h[m] * q[m]$

- Try to enforce no-ISI condition

$$q[m] * h[m] = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases}$$



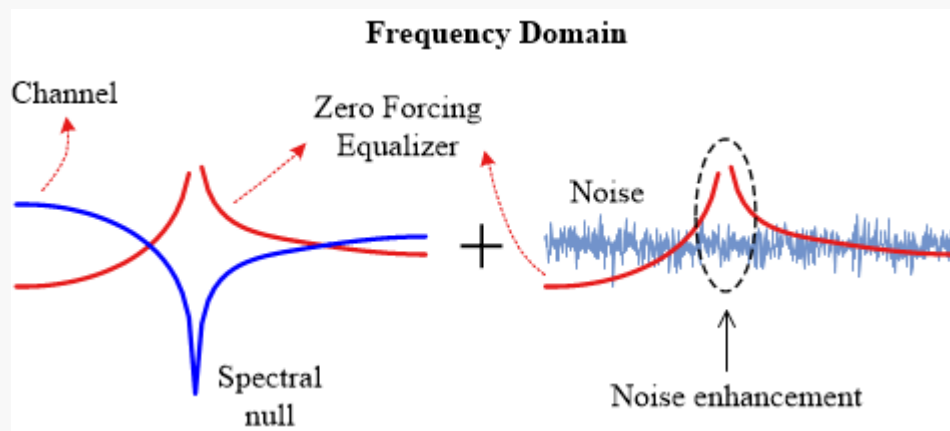
Zero Forcing Equalizer

For an infinitely long equalizer

$$q[m] * h[m] = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases} \longrightarrow Q(F) \cdot H(F) = 1$$

A ZFE simply inverts the channel frequency response $Q(F) = \frac{1}{H(F)}$

Amplifies noise at spectral nulls

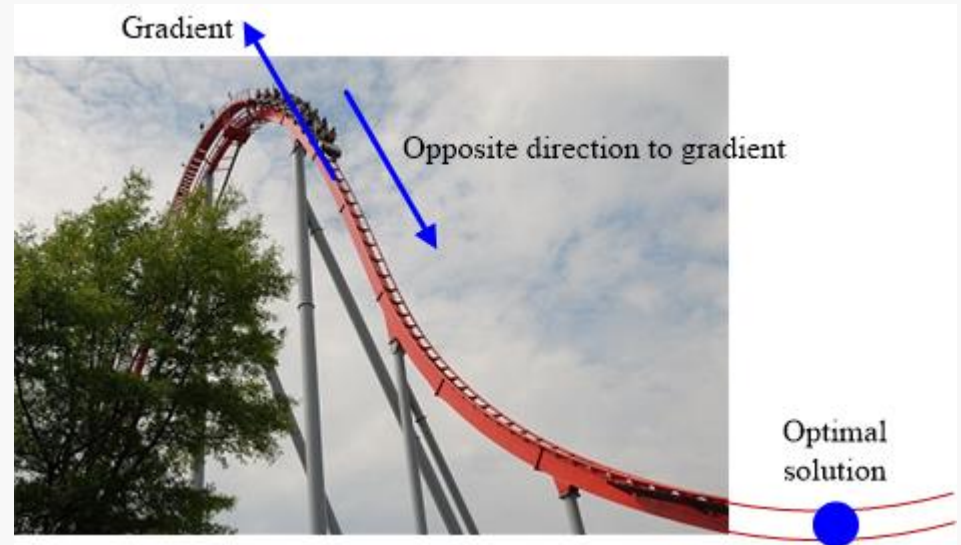


Least Mean Square (LMS) Equalizer

Motivation for an automatic equalizer with self-adjusting taps

Assume that the electricity in a roller coaster goes out

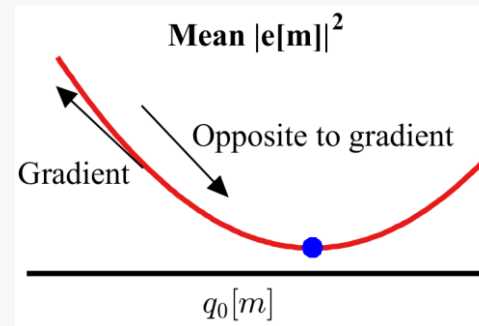
- To prevent the rapid drop, apply the brakes and then leave
- Slowly descend in the direction opposite to the gradient



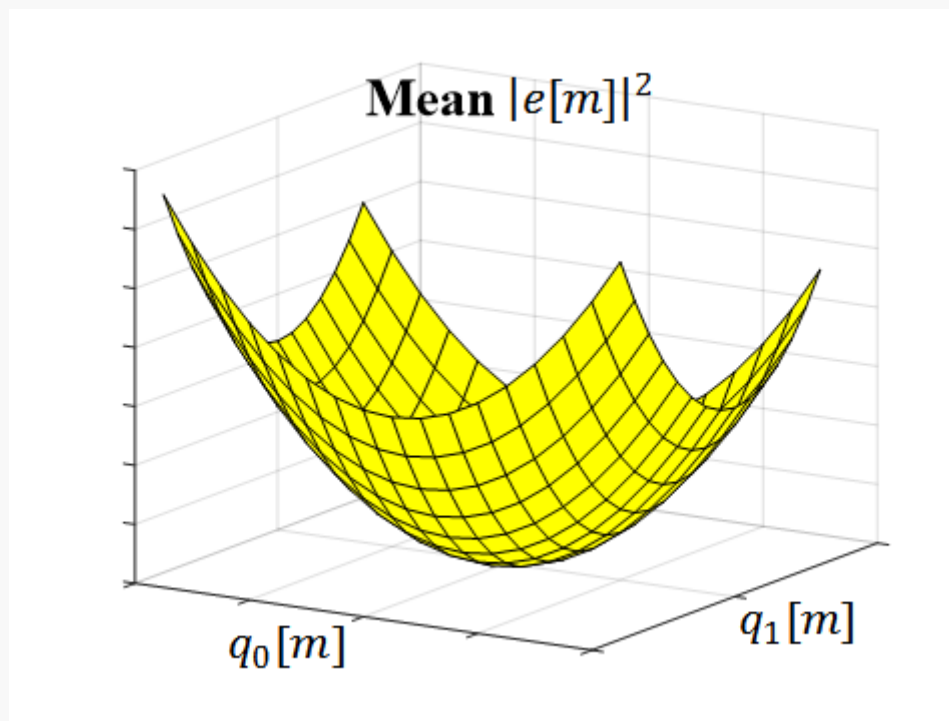
Least Mean Square (LMS) Equalizer

Quadratic is a similar performance function

$$\begin{aligned}\text{Mean}|e[m]|^2 &= \text{Mean}|a[m] - y[m]|^2 \\ &= \text{Mean}|a[m] - z[m] * q[m]|^2\end{aligned}$$



2 tap



1 tap

Least Mean Square (LMS) Equalizer

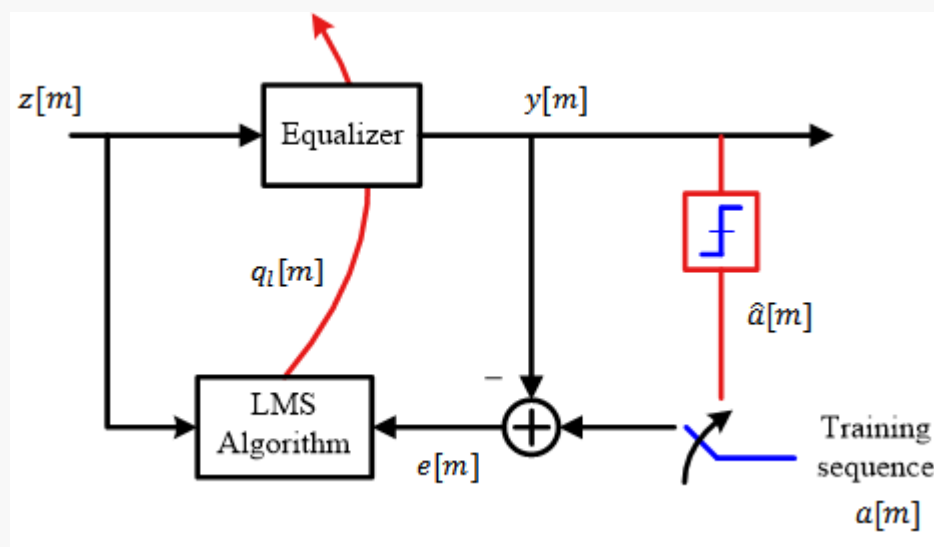
Equalizer update equations are

$$q_l[m+1] = q_l[m] + 2\mu \cdot e[m] \cdot z[m-l]$$

for each $l = -L_q, \dots, L_q$

where μ is the step size

- Training stage
- Decision-directed stage



Least Mean Square (LMS) Equalizer

Example

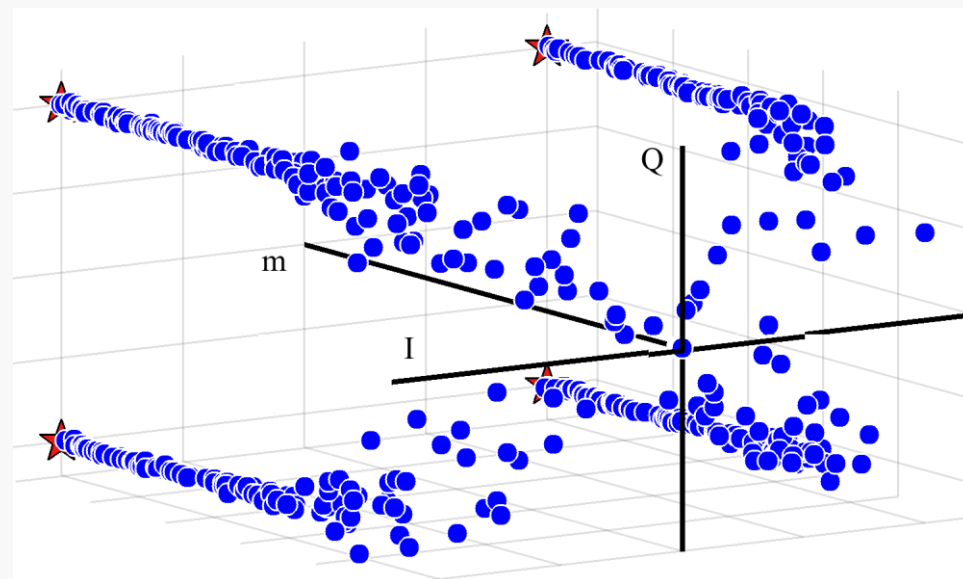
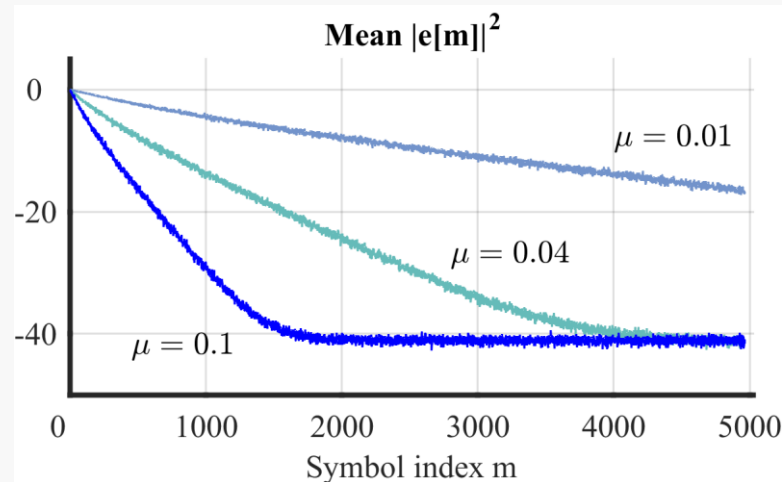
- 4-QAM with excess bandwidth $\alpha = 0.25$

- Channel response

$$h[m] = [0.1 \quad 0.1 \quad 0.67 \quad 0.19 \quad -0.02]$$

- Three different step sizes

$$\mu = 0.01, \quad \mu = 0.04, \quad \mu = 0.1$$



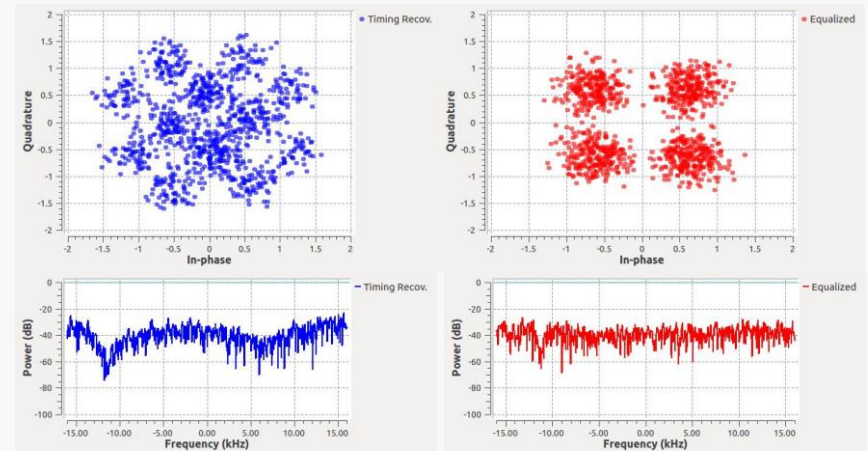
GNU Radio Example

LMS DD Equalizer block

- Gain
- Num. taps
- Samples/symbol
- Constellation object

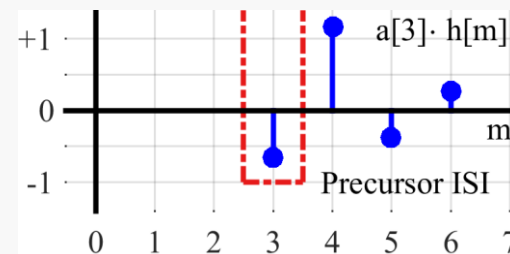
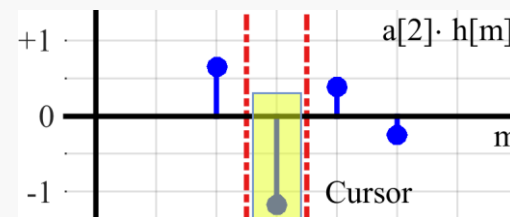
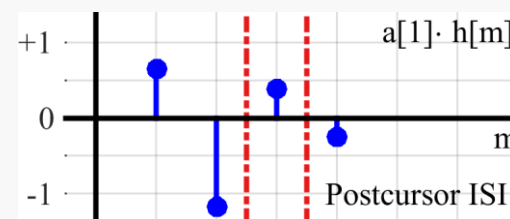
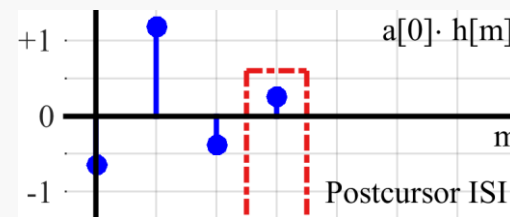
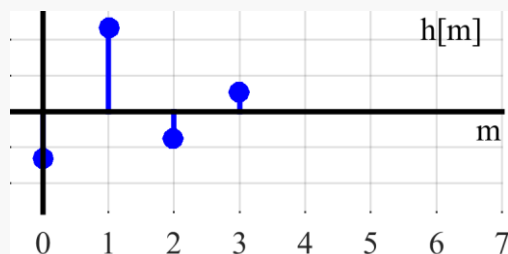
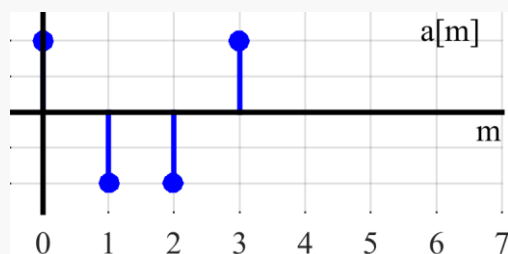
Example: $\mu = 0.01$, 11 taps,

QPSK modulation



Decision Feedback Equalization

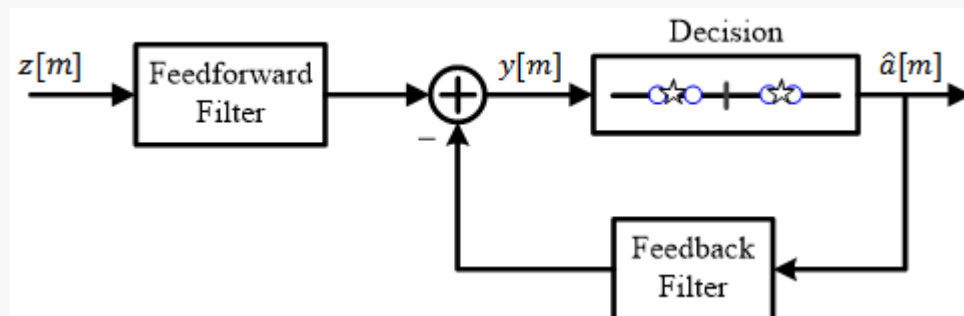
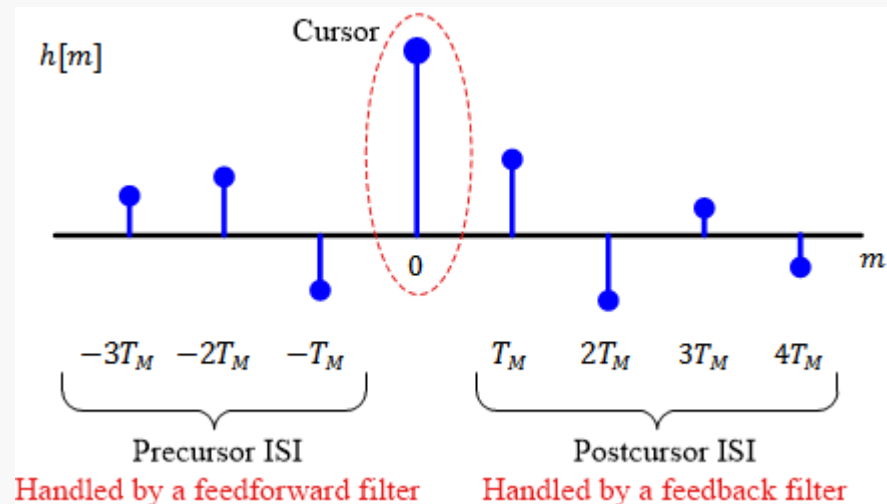
Main idea: decision feedback is possible!



Decision Feedback Equalization

Design strategy

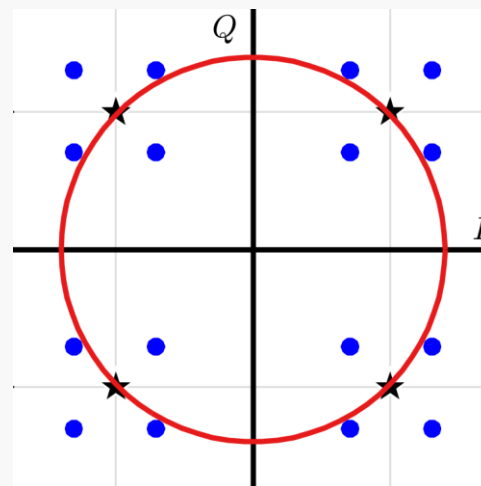
- Split the equalizer in two separate portions
- Feedback filter:** removes the postcursor ISI with the help of symbol decisions
- Feedforward filter:** minimizes the ISI like a regular equalizer



Blind Equalization

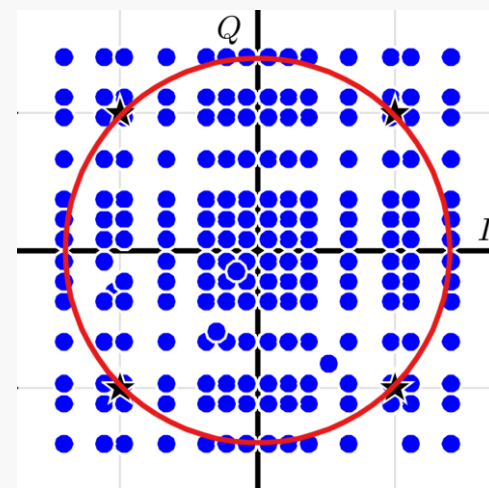
In many situations, no training or decisions are available for channel estimation

There is a structure in the signal, unlike noise



1 tap channel

4 tap channel



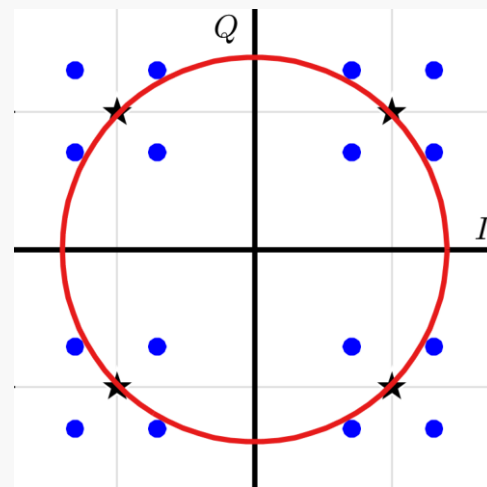
Constant Modulus Algorithm (CMA)

- Assume that Rx samples are available at symbol rate

- For a QPSK constellation

$$\{a_I[m], a_Q[m]\} = \begin{cases} +\frac{1}{\sqrt{2}}, +\frac{1}{\sqrt{2}} \\ +\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}}, +\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \end{cases}$$

- Ring passing through the symbols has a radius of 1
- Distance between the location of actual Rx samples and this ring can be minimized
- Channel is implicitly equalized!

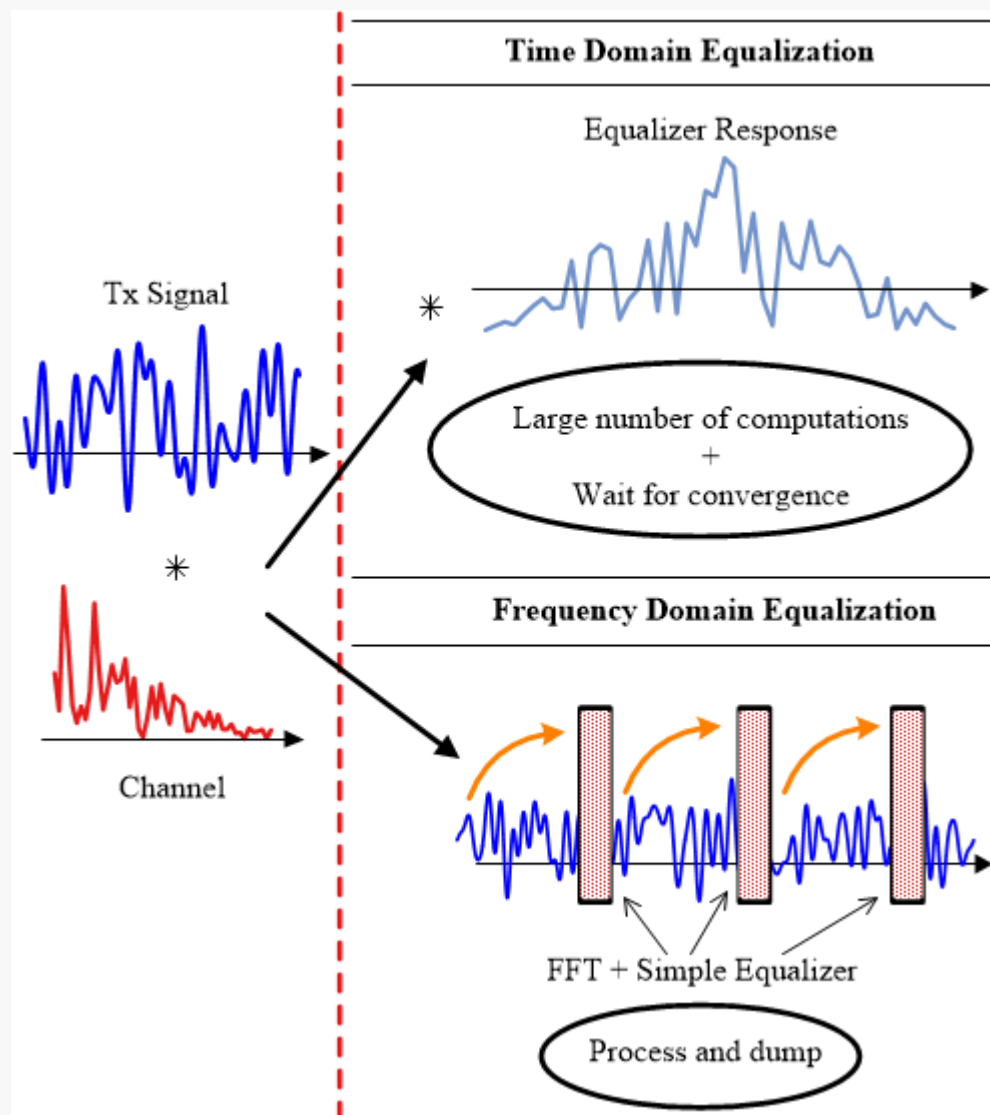


From Time to Frequency Domain

Convergence time (an equalizer is a multi-armed PLL)

Computational complexity

- Convolution of Tx signal with channel is inherently fast (performed by nature)
- Convolution of Rx signal with equalizer taps is inherently slow (done by DSP machine)



Frequency Domain Equalization

Single Carrier – Frequency Domain Equalization (SC-FDE)

- In time domain, at $L = 1$ sample/symbol

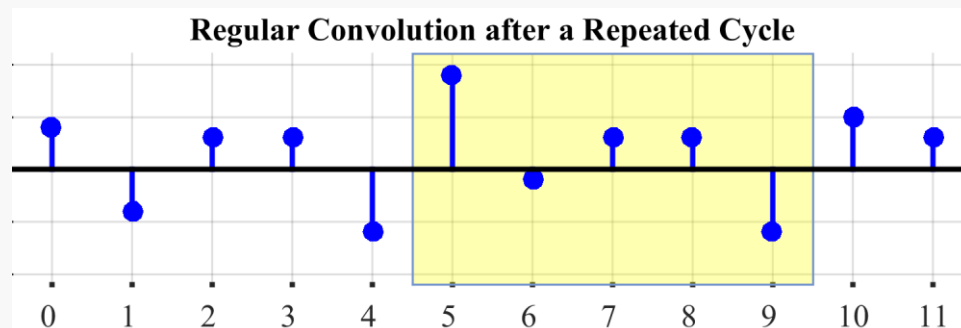
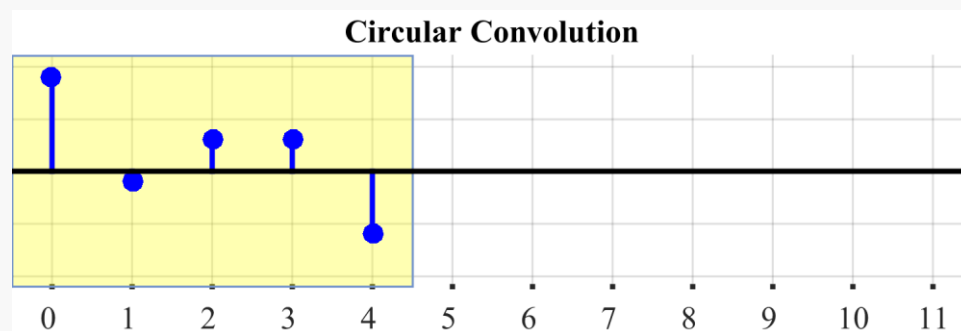
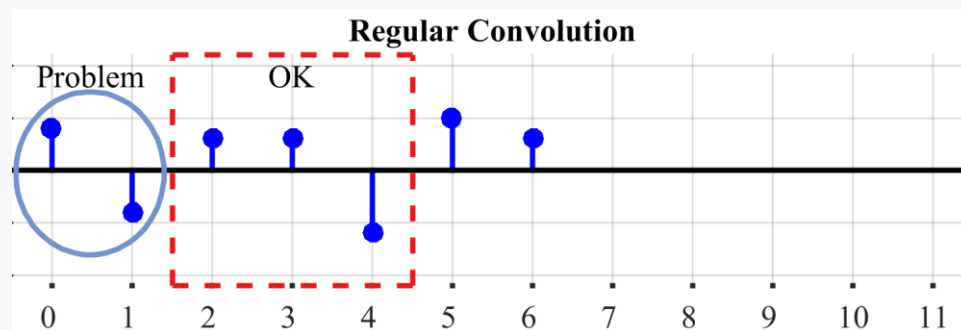
$$z[m] = a[m] * h[m]$$

- In frequency domain, do we have

$$Z[k] \quad =? \quad A[k] \cdot H[k]$$

- This is only true when a circular convolution happens between $a[m]$ and $h[m]$
- **Main question:** *How to make the Rx signal 'look like' having come from a circular convolution procedure when in reality it is not true?*

Frequency Domain Equalization



- Doubling the signal duration is a high price to pay
- Best compromise is to only repeat a portion of the Tx sequence **from the end** with maximum expected channel length
- Known as Cyclic Prefix (CP)

Frequency Domain Equalization

$$Z[k] = A[k] \cdot H[k]$$

- Equalizer operation

$$\hat{A}[k] = \frac{Z[k]}{\hat{H}[k]}$$

- Data symbols recovery

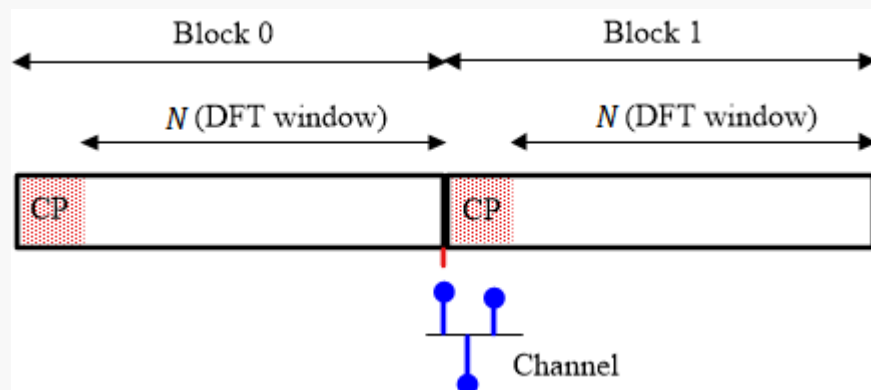
$$\hat{a}[m] = \text{iDFT} \left\{ \hat{A}[k] \right\}$$

Frequency Domain Equalization

We have a second compelling reason for using CP

- Block processing mitigates ISI

Conventional method



Intuitive method

